Savitribai Phule Pune University Fundamentals of Data Structures (210242) SE Computer Engineering
2019 Pattern

Total Marks: 70 | Time: 2½ Hours

Instructions:

1. Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.
4. Assume suitable data if necessary.

Unit III — Searching and Sorting (18 Marks)

Q1) a) Write a pseudo-code for binary search. Apply the algorithm on the following numbers stored in an array to search for the numbers 23 and 100: 9, 17, 23, 40, 45, 52, 58, 80, 85, 95, 100. [9] b) Explain the selection sort algorithm with an algorithm. Sort the following numbers using selection sort and show the contents of the array after every pass: 27, 76, 17, 9, 45, 58, 90, 79, 100. [9]

OR

Q2) a) Explain the quick sort algorithm with a suitable example. What is the time complexity of quick sort? [9] b) Write a short note on sentinel search and indexed sequential search with suitable examples. [9]

Unit IV — Linked List (17 Marks)

Q3) a) Write a pseudo-code to insert a new node into a singly linked list at the beginning, at the end, and at a given position. [9] b) Explain the representation of a polynomial using a Generalized Linked List (GLL) with an example. [8]

OR

Q4) a) What is a doubly linked list? Explain the process of deletion of an element from a doubly linked list with an example. [9] b) What is a dynamic data structure? Explain a circular linked list with its basic operations. [8]

Unit V — Stack (18 Marks)

Q5) a) Write a pseudo-code for the basic operations of a stack (push, pop, peep, and isEmpty) using an array. [8] b) What are the variants of recursion? Explain each variant with an example. [10]

OR

Q6) a) Write an algorithm for postfix expression evaluation. Explain with a suitable example. [8] b) Explain the linked implementation of a stack with suitable example and write operations for push and pop. [10]

Unit VI — Queue (17 Marks)

Q7) a) Write a pseudo-code to implement a circular queue using an array. Explain its basic operations (enqueue, dequeue, isEmpty, isFull). [9] b) Explain the array implementation of a priority queue with all basic operations. [8]

OR

Q8) a) Explain the linked implementation of a queue with suitable example. Write algorithms for enqueue and dequeue operations. [9] b) Write a pseudo-code for insertion and deletion operations of input-restricted and output-restricted double-ended queues (deque). [8]

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